External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Keep the operation ID. Reconcile the state. Authorize any new effect.

Distinguish committed, prepared, unknown and reliably absent outcomes before retry or compensation decisions.

Keep one habit: Status contract → recovery map → printer picture → evidence/authority probes → pure local decision audit → independent unknown-outcome reasoning.

MAIN LESSON · SECTION 1

Recover the operation's state before deciding to repeat it

> A missing acknowledgement leaves uncertainty; it does not erase an effect.

Learning goals

I can explain the **mechanism** and terminology of external side effects and recovery.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve caller timeouts and exact operation identity

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Timeouts retries and cancellation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**
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> _____
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> _____
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> _____
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> _____

End goal

Explain external side effects and recovery using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of external side effects and recovery

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

Keep acknowledgement, committed state and recovery authority distinct

> The returned status must belong to the same operation and an authorized source.

Begin with the lost booking reply

A desk can record a booking successfully while its reply gets lost. Repeating the booking because no reply arrived can create a second effect. Keep the original operation ID and ask an authorized, reliable status source what happened. A committed result can be returned; an unresolved outcome needs reconciliation. Cancelling or otherwise offsetting an earlier effect is another action, with its own approval and record.

Define uncertain external effects and recovery

External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

Complete operation-status and authority contract

Recovery fixture or requirement	Complete local meaning
Original operation ID	BOOK-1, kept stable during recovery

Recovery fixture or requirement	Complete local meaning
Committed fixture	Authoritative matching status; result booking B1 already exists
Prepared fixture	Work was prepared; not proof it failed or is safe to repeat
Unknown/unavailable fixture	Outcome unresolved; review/reconcile rather than assume failure
Reliable absent fixture	Different evidence from silence; still needs separate resend authority and provider duplicate controls
Status access	Trusted permission to inspect the operation is assumed or denied by a supplied fixture flag
Compensation	A separate proposed cancel-booking operation COMP-1, with its own exact approval fixture
Code scope	Pure local decision helper; no status provider, booking, printer or token-transfer call

Name supplied and missing evidence

List the given facts and one fact that is not supplied.

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the existing committed result without creating another effect

> Stable identity and trustworthy status matter more than whether a reply arrived.

Original prepared/committed/unknown policy

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

Complete status-to-decision recovery map

Reconcile the original operation before any new side effect

RELIABLE MATCHING EVIDENCE	RECOVERY DECISION	WHAT MUST NOT HAPPEN
BOOK-1 committed, existing booking B1	Return the existing result to an authorized caller	Do not create a second booking
BOOK-1 prepared	Pending review / inspect authoritative progress	Do not call preparation a confirmed failure
BOOK-1 unknown or status unavailable	Keep outcome unresolved and reconcile	Do not repeat blindly with a fresh key
Status is for another ID or is unreliable	Review the evidence mismatch	Do not report it as BOOK-1's confirmed result
Proposed compensation	Check separate exact authority and preserve original history	Do not silently cancel/delete/refund as part of reading status

The helper starts from supplied status fixtures and creates no effects. A recovery decision and an executed recovery action are separate stages.

State operation identity and evidence assumptions

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

External side effects and recovery — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A booking is committed in storage, but response delivery times out. Caller asks to repeat.	The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action.	Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change the status evidence without inventing permission

> Reliable absence can change the next proposal; it is not automatic approval to act.

Current task and response

- A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict reliably absent printer status

- Change: Status reliably reports J6 absent.

▼ **Check after predicting**

Prediction: A policy-approved resend using the same deduplication key may be considered.

Reason: Confirmed absence supplies different **evidence** from silence.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare an unknown printer outcome with reliable completion or absence

> A changed evidence state does not silently grant a new action.

Main committed-booking scenario and existing-result conclusion

A booking is committed in storage, but response delivery times out. Caller asks to repeat.

Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Keep the reasoning's scope

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate printer job, stable key and complete conditions

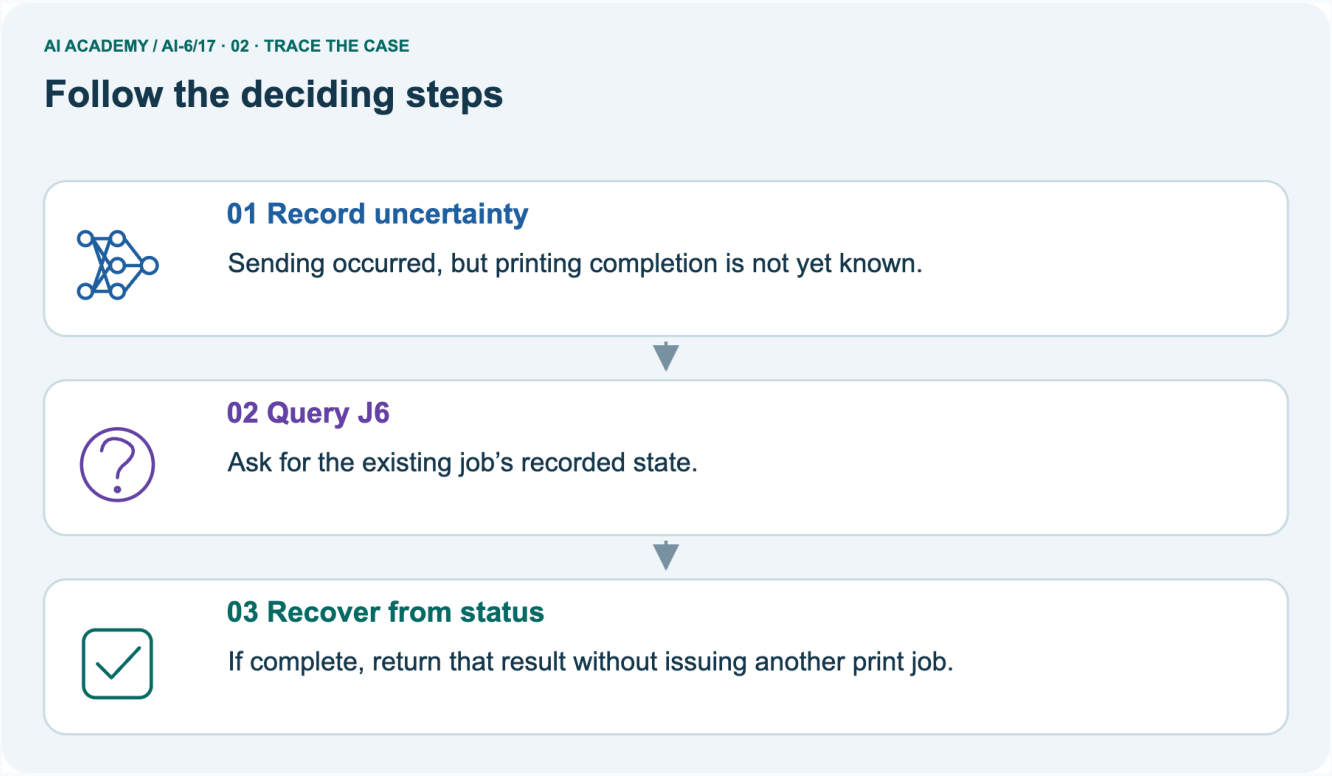
Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

The main booking uses BOOK-1/B1; the picture is a separate printer job J6 for one fictional badge. A missing reply leaves printing outcome unknown. An authorized status lookup that reliably reports J6 complete supports returning the existing result without another print.

Reliable absence is different from silence, but only a separately approved same-key resend with the required provider duplicate controls may be considered. If those controls or reliable status are missing, keep review/uncertainty. No print command is executed in this lesson.

Trace uncertainty, status query and completion recovery



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Record uncertainty	Sending occurred, but printing completion is not yet known.
2. Query J6	Ask for the existing job’s recorded state.
3. Recover from status	If complete, return that result without issuing another print job.

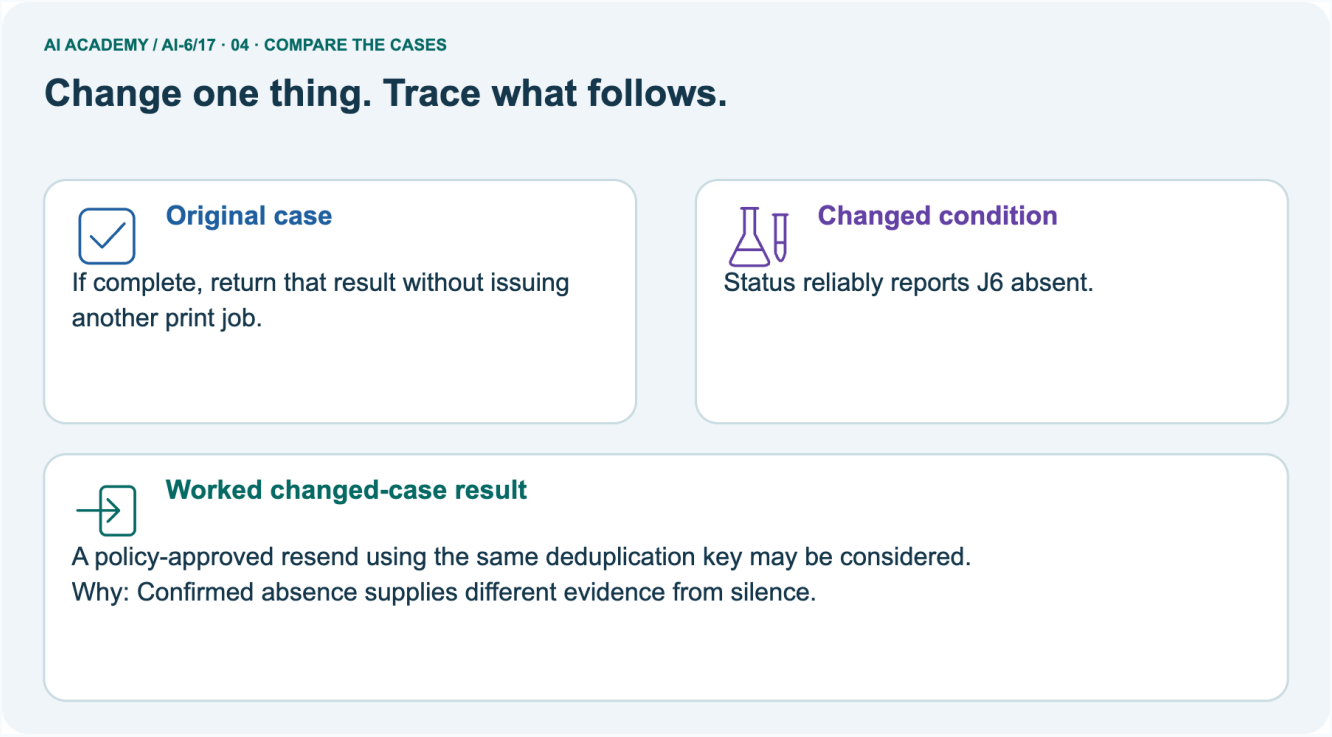
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Predict reliably reported J6 absence

- Change one thing: Status reliably reports J6 absent. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the changed evidence and remaining guarantee limits

- Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

- Read the labelled picture: Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

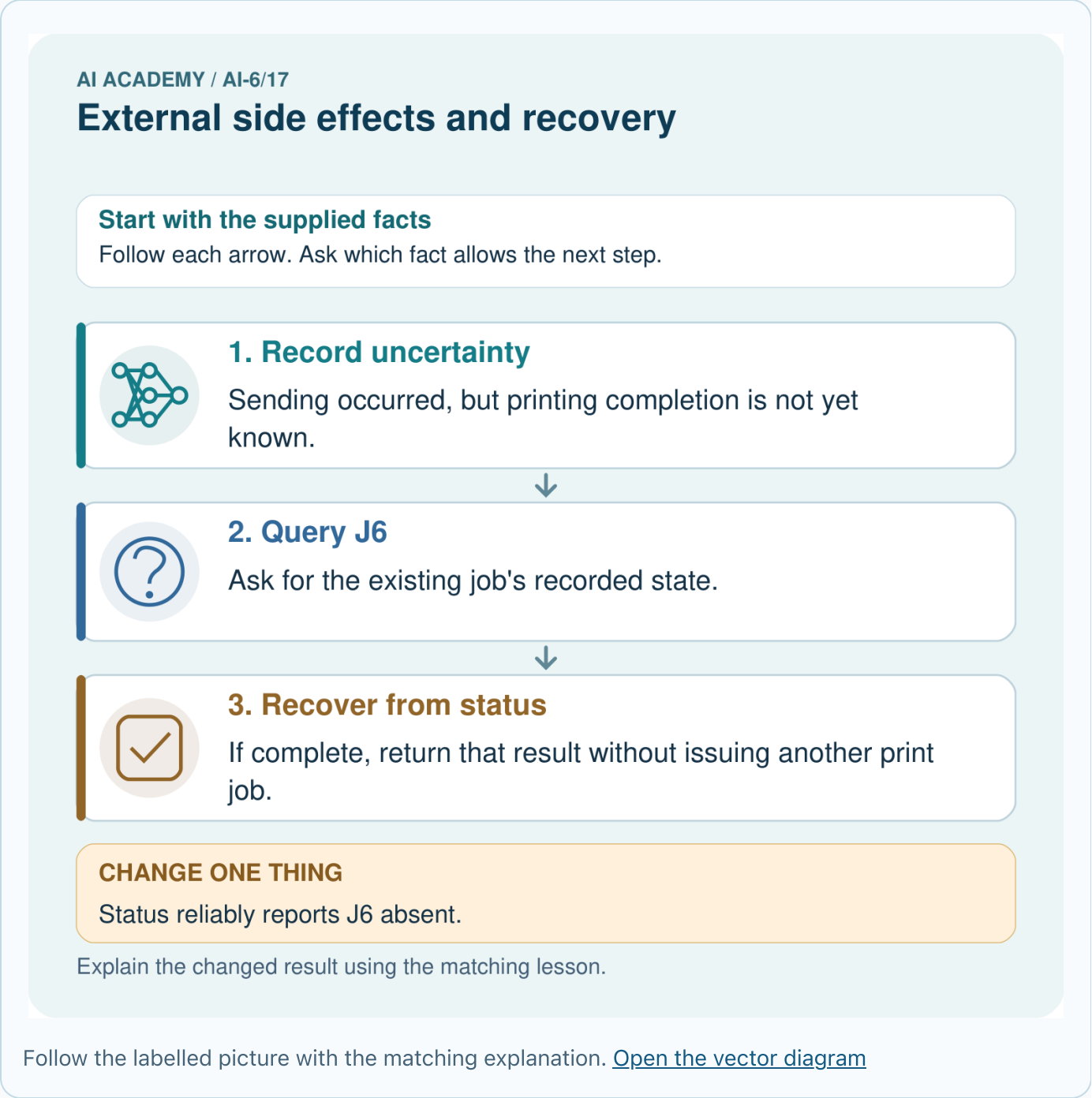
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: External side effects and recovery

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Check evidence, authority and duplicate controls separately

> Compensation is another side effect, not a free consequence of uncertainty.

Core separate-compensation-authority qualification

A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete status, read-access and resend-control probes

Recovery probe	Expected decision or limit
Committed matching status with a valid result	Return existing copied result, no new effect
Prepared	Pending review; do not infer confirmed non-execution
Unknown or unavailable	Reconcile, no blind repeat
Reliable absence but no resend approval	Review; evidence does not authorize the action
Resend approval but unreliable status	Review; permission does not supply missing evidence
Reliable absence + trusted resend approval + provider deduplication	Return a same-key resend candidate, not evidence it was executed
No status-read permission	Deny disclosure under the supplied policy fixture
Compensation requested without its own authority	Reject that separate action; preserve original committed record

Recovery probe	Expected decision or limit
Pure local recovery checks pass	No real provider consistency, durable recovery or exactly-once guarantee established

Guided explanation and repair

- A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

Practise naming what is known and what remains uncertain

> Use the original operation ID and distinguish a decision from an action.

Current explanation and guided scope check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “External side effects and recovery”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual compensation-limit check

- Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Printer picture response

- Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

> **My explanation using a pictured step and the given facts**

Reserve independent classroom-token uncertainty task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

Run the complete status-evidence and recovery-decision audit

> Inspect unchanged original state and explicit conditions before proposing a new effect.

Current lost-booking-acknowledgement investigation

A booking is committed in storage, but response delivery times out. Caller asks to repeat.
Show the intermediate steps and give the conclusion with its scope.

Complete runnable local recovery decisions

```
"""Pure recovery decisions on trusted classroom status fixtures; no effects."""
from copy import deepcopy

def reconcile(operation_id,status_record,*,can_read_status=True,
              resend_approved=False,provider_deduplicates=False):
    if type(operation_id) is not str or not operation_id:
        raise ValueError('operation ID required')
    if any(type(v) is not bool for v in
[can_read_status,resend_approved,provider_deduplicates]):
        raise ValueError('policy flags must be supplied Booleans')
    if not can_read_status:return {'status':'denied'}
    if (type(status_record) is not dict or status_record.get('operation_id')!=operation_id
        or status_record.get('authoritative') is not True):
        return {'status':'review','reason':'missing reliable matching status'}
    state=status_record.get('state')
    if state=='committed':
        result=status_record.get('result')
        if type(result) is not dict:return {'status':'review','reason':'missing committed
result'}
        return
{'status':'return_existing','operation_id':operation_id,'result':deepcopy(result)}
    if state=='prepared':return {'status':'pending_review','reason':'prepared is not
confirmed failed'}
    if state in ('unknown','unavailable'):
        return {'status':'review','reason':'outcome unresolved; do not repeat blindly'}
    if state=='absent':
        if resend_approved and provider_deduplicates:
            return {'status':'resend_candidate','operation_id':operation_id,
                    'requires':'separate authorized execution with same provider key'}
        return {'status':'review','reason':'absence alone does not grant resend authority
or deduplication'}
    return {'status':'review','reason':'unrecognized status'}
```

```
committed={'operation_id':'BOOK-1','authoritative':True,'state':'committed',
           'result':{'booking_id':'B1','status':'booked_fixture'}}
before=deepcopy(committed)
returned=reconcile('BOOK-1',committed)
assert returned['status']=='return_existing' and committed==before
returned['result']['status']='modified caller copy'
assert committed==before
print('Main lost-acknowledgement recovery:',reconcile('BOOK-1',committed))
unknown={'operation_id':'BOOK-1','authoritative':True,'state':'unknown'}
prepared={**unknown,'state':'prepared'}
assert reconcile('BOOK-1',unknown)['status']=='review'
assert reconcile('BOOK-1',prepared)['status']=='pending_review'
assert reconcile('BOOK-1',committed,can_read_status=False)=={'status':'denied'}
print('Unknown/prepared:',reconcile('BOOK-1',unknown),reconcile('BOOK-1',prepared))
# Separate pictured printer job, not an actual print command.
complete={'operation_id':'J6','authoritative':True,'state':'committed',
          'result':{'print_job':'one fictional badge'}}
absent={'operation_id':'J6','authoritative':True,'state':'absent'}
assert reconcile('J6',complete)['status']=='return_existing'
assert reconcile('J6',absent)['status']=='review'
candidate=reconcile('J6',absent,resend_approved=True,provider_deduplicates=True)
assert candidate['status']=='resend_candidate' and candidate['operation_id']=='J6'
assert reconcile('J6',
{**absent,'authoritative':False},resend_approved=True,provider_deduplicates=True)
['status']=='review'
print('Printer complete:',reconcile('J6',complete))
print('Reliable absence without/with separate trusted
controls:',reconcile('J6',absent),candidate)
# Compensation is its own proposed action with its own exact fixture approval.
compensation={'operation':'cancel_booking','original_id':'BOOK-
1','new_operation_id':'COMP-1'}
def plan_compensation(proposal,trusted_approval):
    if trusted_approval!=proposal:raise PermissionError('separate exact compensation
approval required')
    return deepcopy(proposal)  # plan only, not execution or deletion
try:plan_compensation(compensation,None)
except PermissionError:pass
else:raise AssertionError('Unapproved compensation planned')
assert plan_compensation(compensation,deepcopy(compensation))==compensation
assert committed==before
print('Compensation remains a separate approved-plan fixture; original commit is
unchanged.')
print('No real status lookup, resend, print, booking, cancellation, token transfer or
durable recovery is performed.')
```

Expected unchanged records, conditional proposals and limits

1. Predict the decision for each supplied status before running or tracing. The main fixture already contains a committed booking; the helper returns a copied result without changing that record. Prepared and unknown remain unresolved. For J6, reliable absence becomes only a resend candidate when separate supplied authority and provider-deduplication conditions are true; unreliable absence still reviews.

   The compensation helper checks a separately supplied exact approval fixture and returns a plan, not an executed cancellation. Boolean flags and dictionary equality here are trusted classroom assumptions, not real actor/approval verification. No provider call, print, booking, token transfer, refund or durable recovery occurs.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: reason about an unknown classroom-token transfer

> Use the unknown-status evidence and stable operation identity before proposing another action.

Complete independent unknown-outcome task

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

> **My independent procedure, result and limitation**
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save operation identity and reconciliation evidence with each decision

> Keep uncertain outcomes, authority assumptions and separately proposed compensation records.

Project task and artifact

Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a supported state claim and an authorized next proposal

> Unknown does not mean failed, and compensation does not happen automatically.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of external side effects and recovery

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Acknowledgement	A response confirming operation status
Reconciliation	Checking authoritative state after uncertainty
Compensation	A separate action intended to offset a prior effect

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

- A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

Explain which evidence supports your conclusion and which claim remains unproved.

▼ Reference · Check the service reasoning and its limits

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

My limited claim and a useful next check

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

External side effects and recovery — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A booking is committed in storage, but response delivery times out. Caller asks to repeat.	The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action.	Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

By the end, I can	How I will show it
Explain the mechanism and terminology of external side effects and recovery	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Recover an uncertain print job. A badge printer loses its reply after receiving a print request.
- Which labelled step matches this detail? If complete, return that result without issuing another print job.
- 1. Record uncertainty
- 2. Recover from status
- 3. Query J6
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Recover from status If complete, return that result without issuing another print job. A lost response does not show whether an external effect occurred.

Recovery distinguishes unknown outcome from a confirmed failed action.

Changed-case question and optional hint

- Changed-case practice: Status reliably reports J6 absent. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. A policy-approved resend using the same deduplication key may be considered.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Repair and retain the record

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

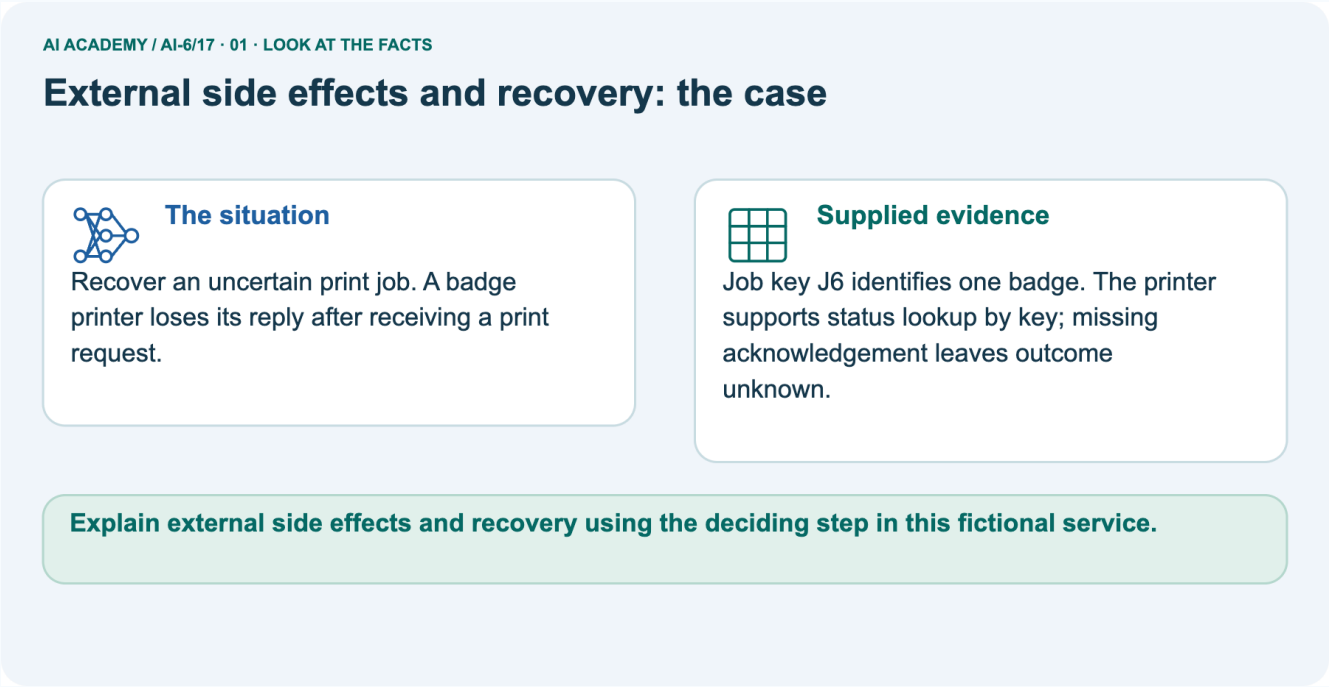
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. External side effects and recovery: the case



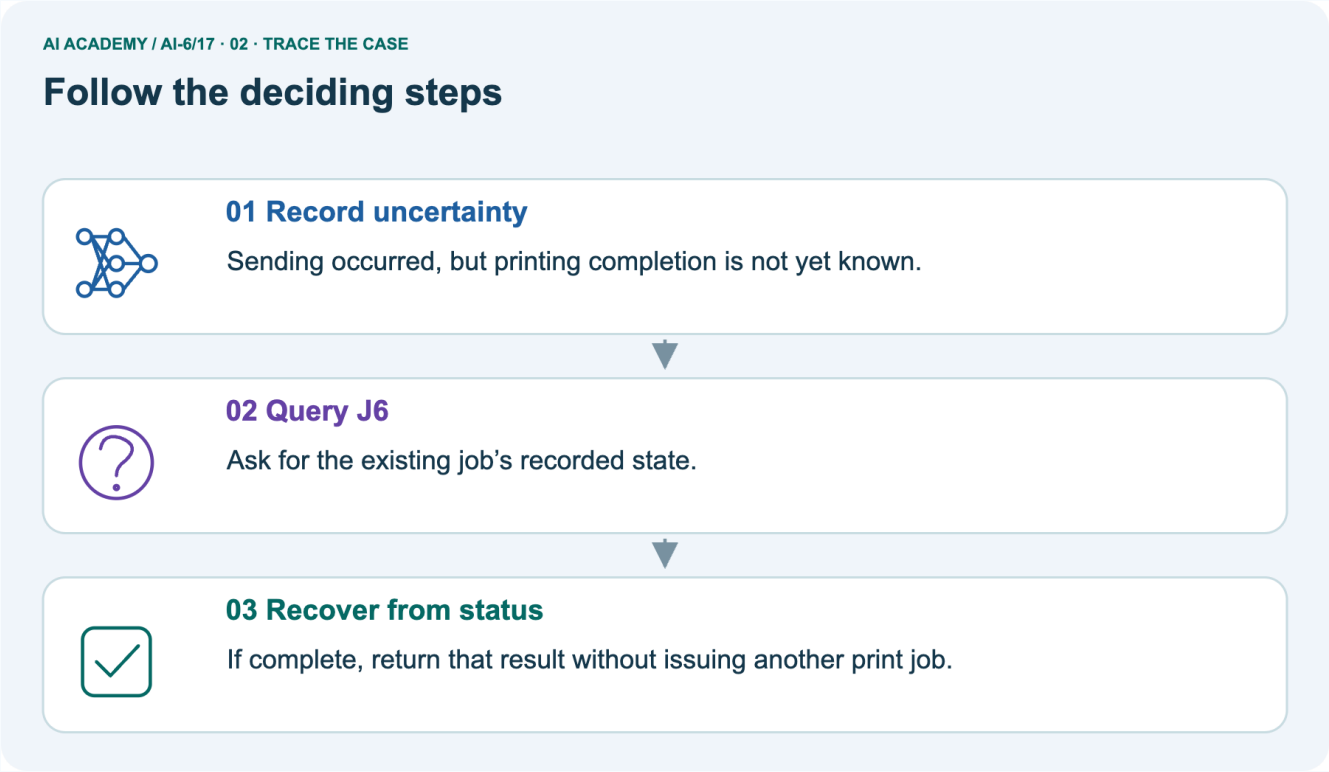
Follow the labelled picture with the matching explanation. Open the vector diagram

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.
Supplied facts: Job key J6 identifies one badge.

The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

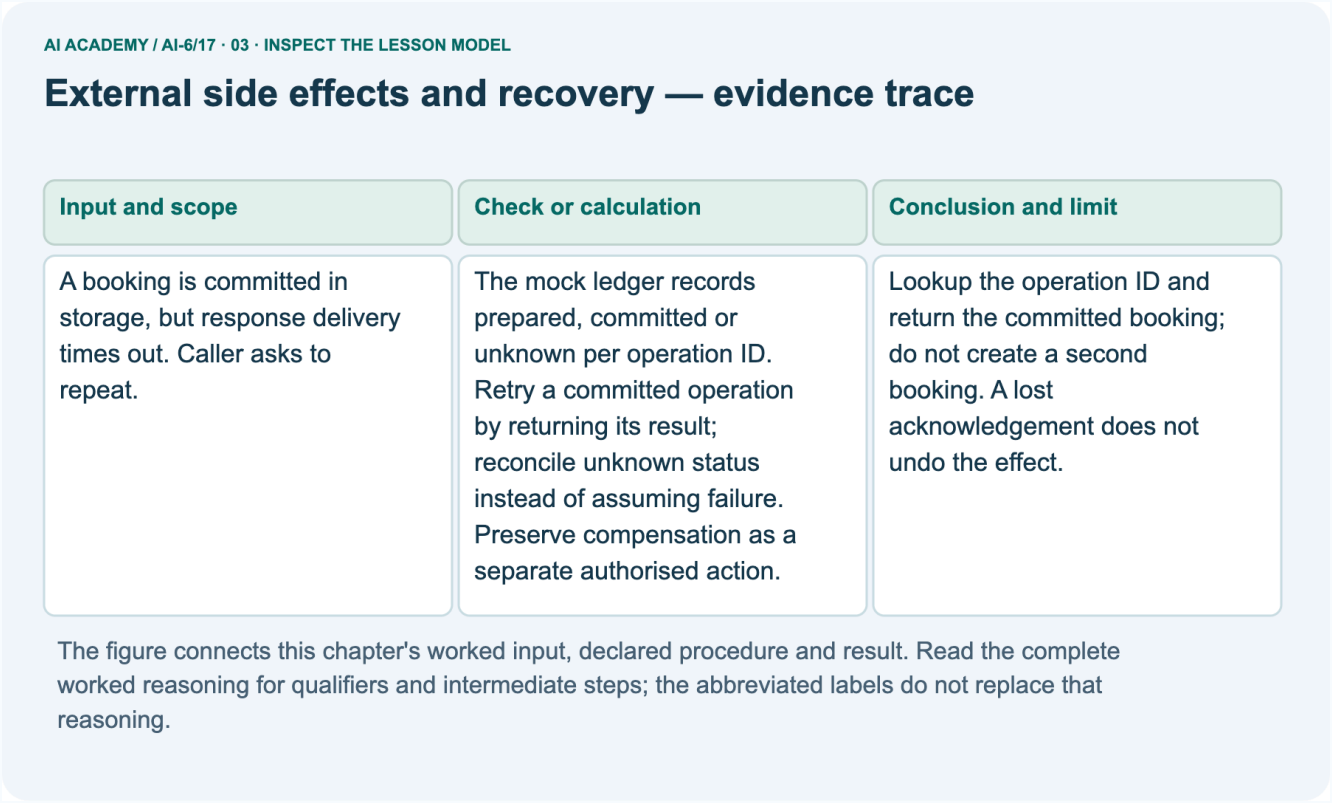


Follow the labelled picture with the matching explanation. Open the vector diagram

Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

Point to each numbered stage and explain which recorded **input** it uses.

03. External side effects and recovery — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

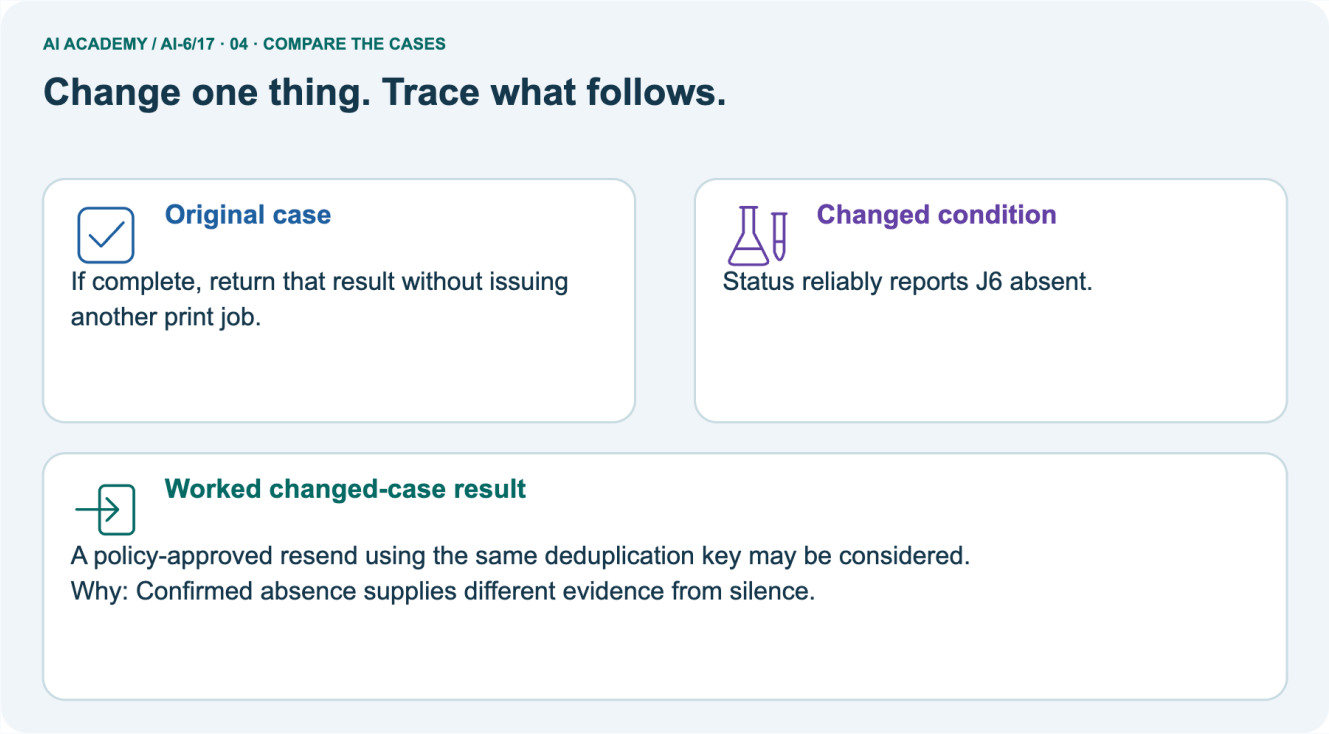
External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



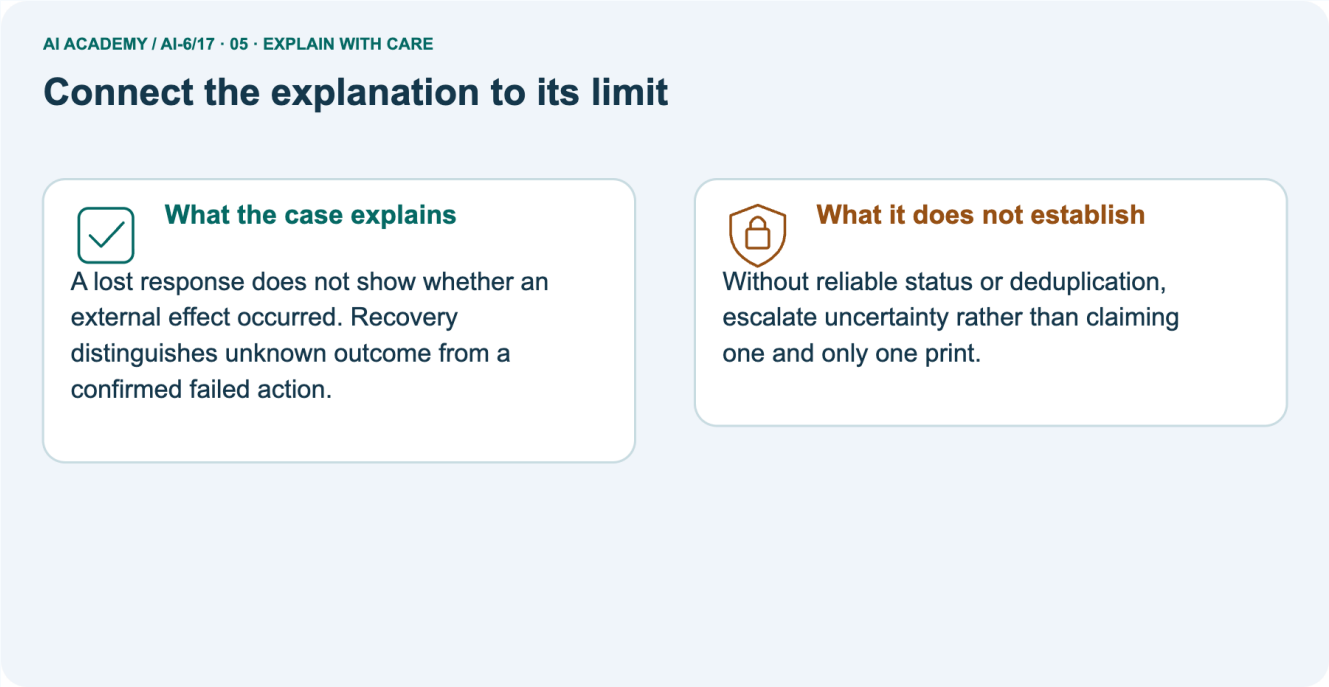
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



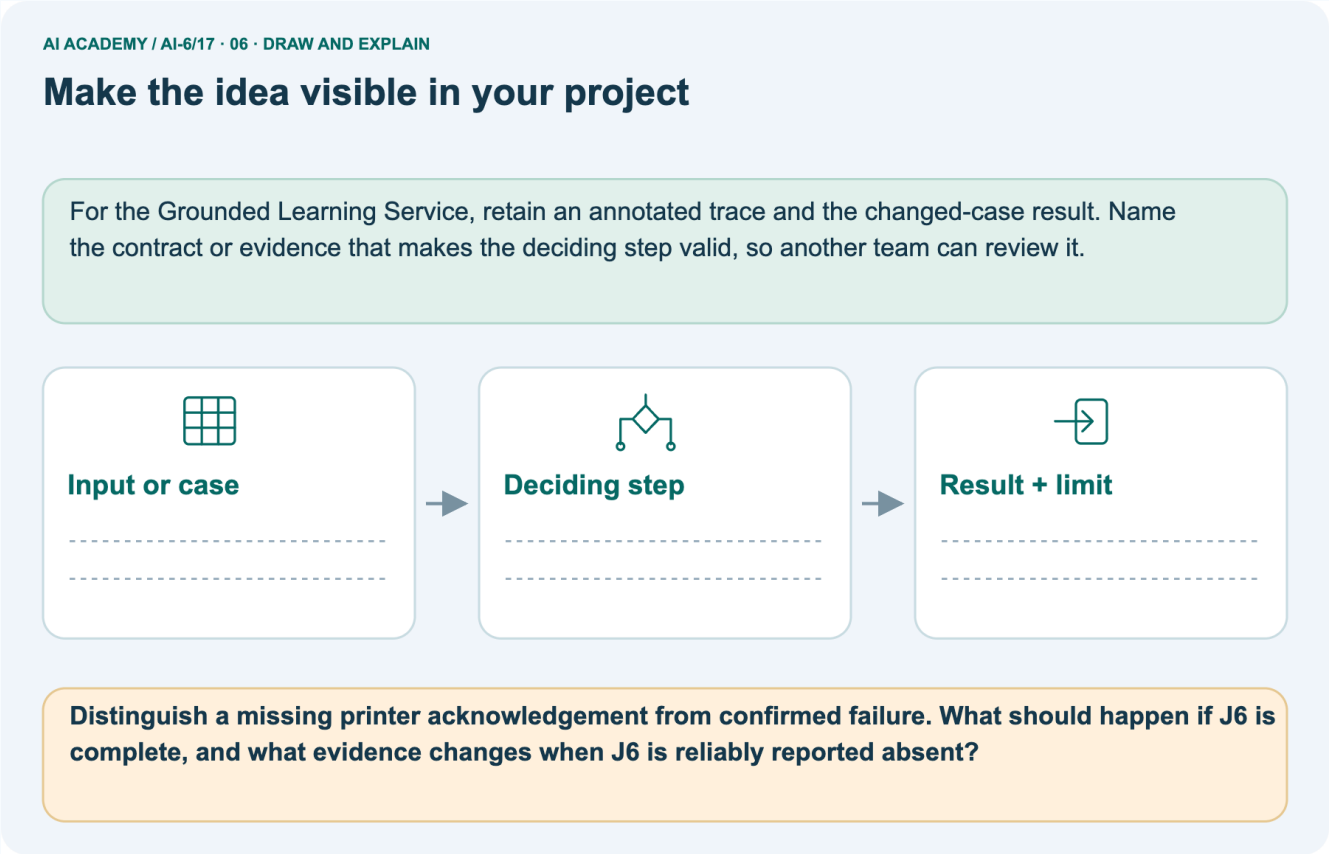
Follow the labelled picture with the matching explanation. Open the vector diagram

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. External side effects and recovery: the case

02. Follow the deciding steps

03. External side effects and recovery — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.